Mohammed Mahdi Jahangiri

Forward Deployed Engineer | AI Engineer

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AI engineer who **embeds LLMs into existing software** and **owns delivery end to end**, from ambiguous requirements to live testing with clients and partner teams. MSc in Artificial Intelligence and 1+ year of professional software delivery, including work across four partner organisations and, as a solo founder, 5+ real-estate agency clients. Brings the same build-it-on-site approach to customer problems: **scope the ambiguity, ship the integration, stay until it works in production**. Fine-tuned a Gemma 4 (2B) that outperformed a Gemma 4 (26B) zero-shot baseline on task routing (**53.4% vs 29.0%**) at sub-second latency.

Technical Skills

Languages: Python, C++, C#, SQL
LLMs & AI: Cloud & local LLM integration, Agent architectures, Structured tool calling, Guardrails, Fine-tuning (QLoRA), Model evaluation, PyTorch, Hugging Face, RAG
Integration & Delivery: FastAPI, REST APIs, JSON Schema, Telemetry, Git, Production debugging
Customer & Delivery: Technical discovery, requirements scoping, stakeholder demos & live test sessions, cross-team coordination, sprint planning (Agile/Scrum), code review, technical documentation

Experience, Projects & Publications

R&D Technical Artist (Contract) — Awkward Studios / Media Cymru
Apr 2026 – Jun 2026

Cardiff — Forward-deployed R&D and client delivery for CULTVR Lab's 360° dome installation.

- Collaborated across four partner organisations, **translating stakeholder requirements into delivered work** and supporting the project through to a **live public QA session**
- With no off-the-shelf specification available, **scoped the problem under ambiguous requirements** and built a custom C# component that corrected UI distortion on the 360° dome projection, **removing a key technical blocker**
- Managed contributions through a GitHub pull-request workflow and travelled regularly to the venue to **test, debug and validate builds** on the physical dome

MSc Dissertation — LLM Agent Integration, Skyrim
Python, REST APIs, JSON Schema, QLoRA

- Architected **Python agent middleware** connecting an existing C++ application to **cloud-hosted and local LLM endpoints** through authenticated REST APIs; implemented **structured tool calls** for dialogue routing, clue disclosure and quest-state actions, with **JSON Schema guardrails** blocking invalid actions before execution
- Fine-tuned a local 2B model using **QLoRA** on consumer hardware, increasing task-routing accuracy from **19.0% to 53.4%** and outperforming a 26B zero-shot baseline (29.0%) at **0.98-second latency**
- Built an **automated evaluation and telemetry pipeline** covering 1,500 trials and measuring routing accuracy, output compliance, latency and token usage; fine-tuning reduced raw clue-policy violations from 25 to 5, while runtime guardrails ensured **zero invalid state changes reached the game**
- **Open-sourced the full project** with comprehensive documentation on GitHub and published the model on Hugging Face

Engineering Team Coordinator — Landell Games
Feb 2025 – Sep 2026

Remote / Sweden — Promoted from Engineering Intern (Dec 2024 – Jan 2025). Technical lead of the engineering team.

- Owned the **weekly build and release process**, conducted code reviews and managed the shared Git repository
- Led a **3–6 person team**, assigning tasks through an **Agile sprint workflow** and raising sprint success from **40% to 90%**
- **Profiled and debugged performance bottlenecks**, increasing frame rate from approximately 10 FPS to 30 FPS
- Mediated between engineering and other departments, resolving **cross-disciplinary ambiguities into actionable tasks**

ChwedlAR — DiffTrailblazers 1st Place (£1,500), Cardiff University
AR, RAG, conversational AI

- Served as **team lead and lead engineer**, coordinating development and delivering an award-winning AR museum prototype
- Built a prototype where visitors talk to historical characters, incorporating **RAG** and character-specific AI models

Voice and Emotion Integration Prototype
Python, FastAPI, C++, Whisper

- Built a local **FastAPI service with custom REST endpoints** for real-time communication between Python inference components and an external C++ application
- Integrated Whisper, Wit.ai and a locally trained Hugging Face classifier for **transcription, intent recognition and emotion classification**
- Transmitted **structured inference results** into Unreal Engine 5 behaviour-tree logic to control runtime application behaviour

Additional Projects & Publication

MetaHomeG: Built a Meta Quest VR app, translating requirements from **5+ real-estate agencies** into product features
Meta Quest Store: Shipped a Unity/C# game simulation, owning architecture, compliance submission and official store release
ICHORA 2024: Published and presented a resource-efficient local LLM dialogue pipeline balancing quality and compute cost

Education

MSc Artificial Intelligence — Cardiff University
2025–2026

Dissertation: Locally hosted LLM-driven NPC architecture with deterministic runtime guardrails.

- **Computer Vision & NLP:** RF-DETR object localisation; pandas/scikit-learn pipelines for auto labelling and classification
- **AI Safety Founding Fellow (Spring 2026):** Completed intensive curriculum covering model evaluation methodologies, catastrophic risk mitigation, and empirical AI governance

BEng Computer Engineering — Azad University, Tehran
2020–2025

- **First-Class Honours**; core modules: Data Structures & Algorithms, C++, SQL & Database Design, Networks